

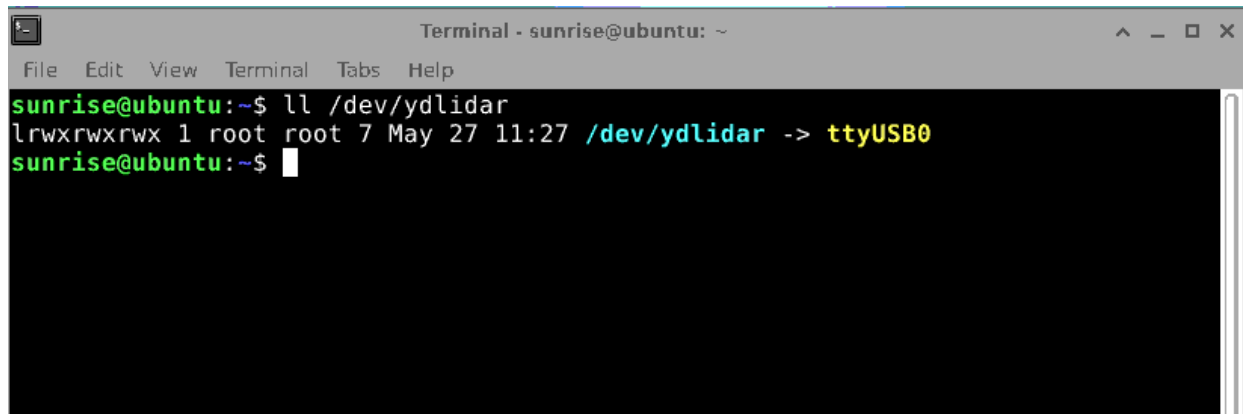
# T-MINI PLUS LiDAR Driver

## 1、 Switch to the Correct LiDAR Model

First, check whether the device is detected. Run the following in the terminal:

```
ll /dev/ydlidar
```

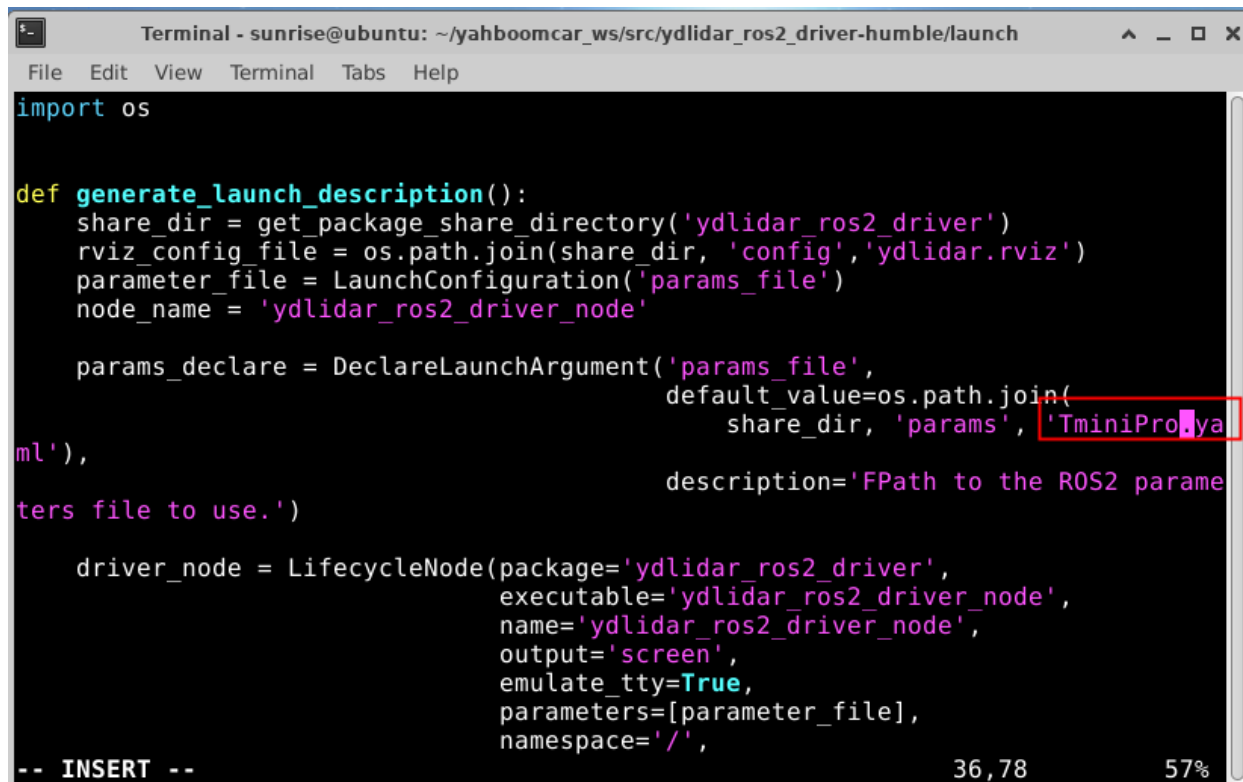
If the device is recognized, the output will appear as shown in the screenshot.



```
Terminal - sunrise@ubuntu: ~
File Edit View Terminal Tabs Help
sunrise@ubuntu:~$ ll /dev/ydlidar
lrwxrwxrwx 1 root root 7 May 27 11:27 /dev/ydlidar -> ttyUSB0
sunrise@ubuntu:~$
```

Next, modify the corresponding YAML file. To drive the T-MINI LiDAR, run:

```
vi yahboomcar_ws/src/ydlidar_ros2_driver-humble/launch/ydlidar_launch_view.py
```



```
Terminal - sunrise@ubuntu: ~/yahboomcar_ws/src/ydlidar_ros2_driver-humble/launch
File Edit View Terminal Tabs Help
import os

def generate_launch_description():
    share_dir = get_package_share_directory('ydlidar_ros2_driver')
    rviz_config_file = os.path.join(share_dir, 'config', 'ydlidar.rviz')
    parameter_file = LaunchConfiguration('params_file')
    node_name = 'ydlidar_ros2_driver_node'

    params_declare = DeclareLaunchArgument('params_file',
                                           default_value=os.path.join(
                                           share_dir, 'params', 'TminiPro.yaml'),
                                           description='Path to the ROS2 parameters file to use.')

    driver_node = LifecycleNode(package='ydlidar_ros2_driver',
                                executable='ydlidar_ros2_driver_node',
                                name='ydlidar_ros2_driver_node',
                                output='screen',
                                emulate_tty=True,
                                parameters=[parameter_file],
                                namespace='/')

-- INSERT --
36,78 57%
```

vi Editor Tips: After opening the file, press **i** to enter edit mode. When done editing, press **ESC**, then type **:wq** to save and exit.

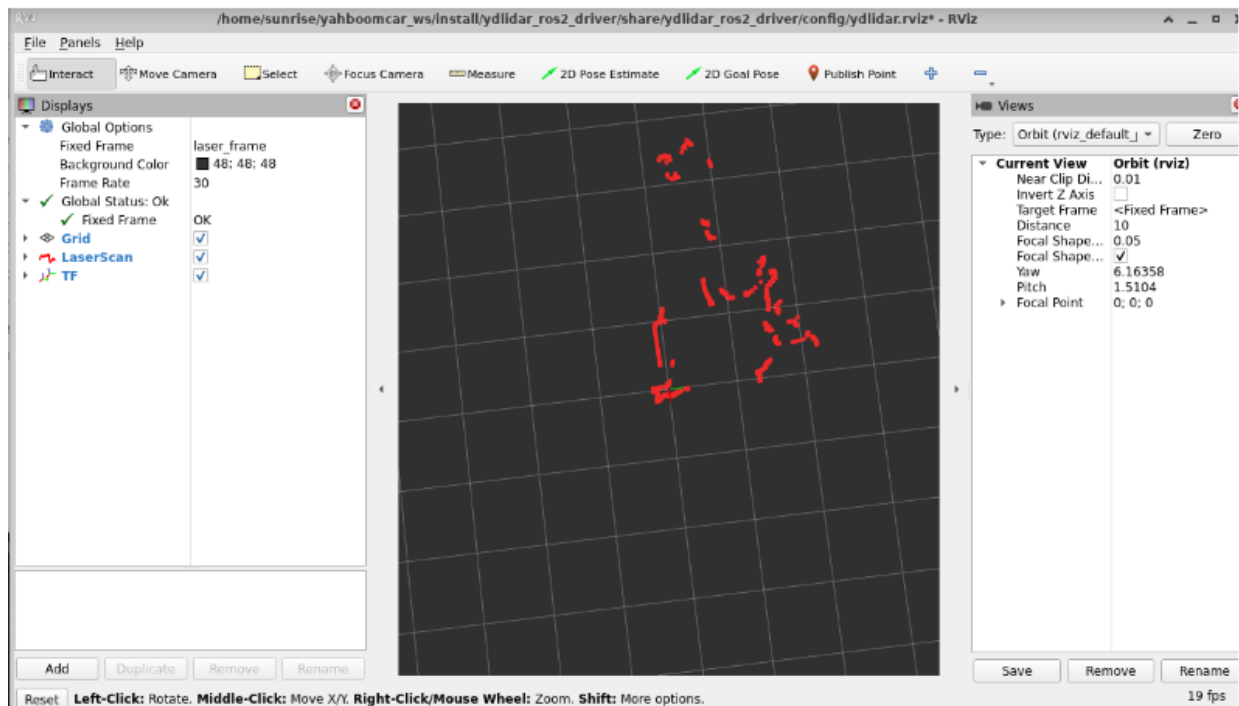
Change the YAML filename to uppercase **TminiPro** — this will allow the driver to launch the TminiPro LiDAR successfully. After making the change, save and exit. Then rebuild the workspace:

```
cd ~/yahboomcar_ws/  
colcon build
```

## 2、 Launch the LiDAR

Open a terminal and enter the following command to start the LiDAR and display its output in RViz:

```
ros2 launch ydlidar_ros2_driver ydlidar_launch_view.py
```



To view the LiDAR node data via topic, run:

```
ros2 topic echo /scan
```

```
header:
  stamp:
    sec: 1716550083
    nanosec: 60639000
  frame_id: laser_frame
angle_min: -3.1415927410125732
angle_max: 3.1415927410125732
angle_increment: 0.0031120283529162407
time_increment: 4.928494672640227e-05
scan_time: 0.10083699971437454
range_min: 0.00999999776482582
range_max: 50.0
ranges:
- 0.0
- 1.0700000524520874
- 1.065999984741211
- 1.0640000104904175
- 1.0670000314712524
- 1.062999963760376
- 1.0540000200271606
- 1.0540000200271606
- 1.0410000085830688
- 1.0449999570846558
- 1.0449999570846558
- 1.0449999570846558
- 1.0449999570846558
- 1.0390000343322754
- 1.0360000408721024
```